

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
varnikakomali@VARNIKAs-MacBook-Air ~ % cd OS_Lab
varnikakomali@VARNIKAs-MacBook-Air OS_Lab % cd Practical
varnikakomali@VARNIKAs-MacBook-Air Practical % nano practical17.c
varnikakomali@VARNIKAs-MacBook-Air Practical % gcc practical17.c -o practical17 && ./practical17
```

```
=== Process Address Space Layout (PID: 30915) ===
```

```
Code / Text Segment:      0x104c30548
Data Segment (Init):      0x104c38000
BSS Segment (Uninit):      0x104c38008
Data/BSS (Static):        0x104c38004
Heap Segment:              0x1054d98e0
Stack Segment:             0x16b1cf1f0
```

```
=== System Virtual Memory Map Summary ===
```

Stack	8176K	32K	32K	0K	0K	0K	0K	1
__DATA	206K	173K	58K	0K	0K	0K	0K	34
__DATA_CONST	106K	106K	0K	0K	0K	0K	0K	43
__DATA_DIRTY	153K	153K	132K	0K	0K	0K	0K	35
__TEXT	6172K	5916K	0K	0K	0K	0K	0K	46
unused but dirty shlib __DATA	58K	58K	58K	0K	0K	0K	0K	19

```
varnikakomali@VARNIKAs-MacBook-Air Practical %
varnikakomali@VARNIKAs-MacBook-Air Practical %
```